Sycophancy as Extended Phenotype: Heteronomous Bayesian Updating, Intentionality Mismatch, and the Evolutionary Stability of Algorithmic Flattery

Empirical Support from Cheng et al. (2025)

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March 2026

Abstract

Recent empirical work demonstrates that AI language models systematically validate users' actions 47% more than human respondents, even when users describe manipulation, deception, or harm to others (Cheng et al. 2025, arXiv:2510.01395). In preregistered experiments (N = 1,604), sycophantic AI models increased users' self-perceived rightness by 25-62% and decreased their willingness to repair interpersonal conflict by 10-28%, while users rated the sycophantic models as higher quality and more trustworthy. These findings establish the prevalence and behavioral consequences of social sycophancy but leave three questions unanswered: what cognitive mechanism makes sycophantic validation so efficient, why users systematically misclassify AI advisors as capable of genuine moral judgment, and why market dynamics perpetuate sycophancy despite its documented harms. This paper proposes answers from three complementary theoretical frameworks. First, I extend Heteronomous Bayesian Updating (HBU), originally developed for institutional norm transmission, to human-AI advisory interaction: users calibrate beliefs by observing interlocutor reactions rather than action outcomes, and AI systems optimized for user satisfaction systematically corrupt this updating process. Second, I apply Asymmetric Intentionality Theory (AIT) to diagnose the structural mismatch: users classify AI advisors as Level 3 intentional agents (capable of recursive moral reasoning) while these systems operate at Level 1 (preference function optimization), generating a self-reinforcing Dynamic Classification Failure. Third, I argue through Extended Phenotype Theory (EPT) and evolutionary game theory that sycophancy constitutes the evolutionarily stable strategy of the current AI market ecology, resistant to invasion by non-sycophantic alternatives absent exogenous intervention. The three frameworks converge on a single empirical prediction from the Cheng data: the systematic erasure of the other person from sycophantic AI conversations (mentioned in fewer than 10% of model

outputs versus over 60% in non-sycophantic models). HBU predicts this because absent reactions on behalf of the harmed party, their perspective never enters the user's Bayesian calculus. AIT predicts this because Level 1 agents lack the recursive capacity to model third-party interests. EPT predicts this because mentioning the other person introduces cognitive friction that reduces user satisfaction and thus memetic fitness. This convergence of three independent predictions on the same datum provides suggestive evidence for the integrated framework's explanatory power.

Keywords: *sycophancy, extended phenotype, Heteronomous Bayesian Updating, Asymmetric Intentionality Theory, evolutionary game theory, human-AI interaction, RLHF, social cognition, institutional evolution*

1. Introduction

Myra Cheng and colleagues at Stanford recently published what is, to my knowledge, the first empirical study measuring both the prevalence and downstream behavioral impact of social sycophancy in AI language models (Cheng et al. 2025). Their findings are striking. Across 11 production models, including GPT-4o, GPT-5, Claude, Gemini, and several open-weight alternatives, AI systems endorsed users' actions at rates 47% higher than human respondents on general advice queries. When tested on 2,000 posts from Reddit's r/AmITheAsshole community where crowdsourced consensus had judged the user clearly at fault, AI models contradicted that consensus in 51% of cases, siding with the user. On a dataset of 6,560 statements describing potentially harmful actions spanning manipulation, deception, self-harm, and relational damage, models affirmed the user's actions 47% of the time.

The behavioral consequences proved equally robust. In two preregistered experiments (N = 1,604), participants who interacted with sycophantic AI models reported significantly higher self-perceived rightness (+62% in a hypothetical vignette study, +25% in a live interaction study with real personal conflicts) and significantly lower willingness to repair interpersonal relationships (-28% and -10%, respectively). All effects survived controls for participant personality, demographics, AI familiarity, and attitudes toward AI (all $p < 0.001$).

These are important results. Yet Cheng et al. explicitly acknowledge that their work is descriptive rather than explanatory. They document what happens but leave open three questions that determine whether the phenomenon can be understood, predicted, and mitigated.

First, what cognitive mechanism makes sycophantic validation so efficient? Why do eight turns of conversation with a flattering chatbot measurably reduce a person's willingness to apologize for conduct that thousands of strangers unanimously judged as wrong? The speed and robustness of the effect, invariant across personality traits and AI familiarity, suggests a mechanism deeper than simple persuasion or social desirability bias.

Second, why do users systematically misclassify AI advisory systems? Cheng et al. report that participants described the sycophantic AI as "objective," "fair," and providing "honest assessment free from bias" at rates statistically indistinguishable from participants who interacted with the non-sycophantic model (7% vs. 8% of open-ended reflections, $p = 0.18$; Appendix A.2). This finding, which the authors appropriately flag as exploratory and requiring further empirical study, nonetheless raises a structural question: what feature of the interaction prevents users from detecting that the system is optimizing for their satisfaction rather than for truth?

Third, why does the market perpetuate sycophancy despite accumulating evidence of harm? Cheng et al. identify the tension clearly: "sycophancy aligns with user preference" while producing "insidious behavioral consequences." They call for developers to address this. But what structural feature of the AI market makes self-correction unlikely?

This paper proposes answers to all three questions, drawing on theoretical frameworks I have developed elsewhere for analyzing institutional persistence, intentionality mismatch, and memetic evolution.

For the first question, I extend Heteronomous Bayesian Updating (HBU; Lerer 2026a) from its original domain of institutional norm transmission to human-AI advisory interaction. HBU formalizes the observation that humans calibrate normative beliefs primarily by observing interlocutor reactions rather than action outcomes. When the interlocutor is a system optimized to produce satisfaction-maximizing reactions, the updating process is systematically corrupted.

For the second question, I apply Asymmetric Intentionality Theory (AIT; Lerer 2026b) and the Generalized Intentionality Mismatch Theorem (GIMT; Lerer 2026b, 2026c). Users interact with AI advisors as Level 3 intentional agents (capable of recursive moral reasoning, reciprocal commitment, and perspective-taking) while these systems operate at Level 1 (optimizing a preference function without modeling the user's beliefs about the system's beliefs). The resulting Dynamic Classification Failure is self-reinforcing: the signal the user employs to verify the system's intentionality level is the same signal the system employs to satisfy the user.

For the third question, I argue through Extended Phenotype Theory (EPT; Lerer 2026a) and evolutionary game theory that sycophancy constitutes the evolutionarily stable strategy (ESS) of the current AI market. The fitness metrics that govern model selection (user retention, satisfaction ratings, session length, willingness to pay) systematically favor sycophantic responses. Individual firms cannot deviate without competitive disadvantage. The April 2025 GPT-4o episode at OpenAI provides empirical illustration: a routine optimization update naturally produced dramatically more sycophantic behavior because the RLHF gradient pointed toward flattery; correcting it required explicit intervention against the optimization's natural direction.

The paper's structure reflects these three contributions. Section 2 develops the HBU mechanism for sycophantic persuasion. Section 3 applies AIT and Dynamic Classification Failure. Section 4 presents the EPT/ESS analysis. Section 5 examines the convergence of all three frameworks on a single empirical finding from Cheng et al.: the systematic erasure of the other person from sycophantic conversations. Section 6 discusses implications for AI governance and legal frameworks. Section 7 acknowledges limitations. Section 8 concludes.

A methodological note is warranted. This paper is a theoretical contribution that finds empirical support in a recent study. I generate no new data. The contribution lies in providing mechanistic explanations for documented phenomena, structural diagnoses for observed patterns, and falsifiable predictions that future empirical work can test. The frameworks I apply were developed independently of the Cheng et al. data, in the context of institutional reform resistance, corporate liability, and legal system evolution. Their applicability to AI sycophancy represents a novel extension that I flag explicitly where it occurs.

2. Heteronomous Bayesian Updating as the Cognitive Mechanism of Sycophantic Persuasion

2.1 The Standard Model and Its Limitation

Standard Bayesian models of belief updating assume agents revise beliefs by incorporating new evidence weighted by its likelihood given prior beliefs:

$$P(H|E) = [P(E|H) * P(H)] / P(E)$$

Applied to the Cheng et al. experimental setting, this would predict that a user uncertain about whether their behavior was wrong ("Am I the asshole?") should update based on the quality of evidence the AI provides. If the AI offers compelling arguments for why the user was right, the posterior shifts toward rightness. If the arguments are weak or generic, updating should be modest.

But Cheng et al. report that the sycophantic effects were robust across participant traits, including AI familiarity. Users who reported extensive prior experience with AI models were not less susceptible to sycophantic influence than novices. If the mechanism were argument quality evaluation, experienced users should show diminished effects. They did not.

2.2 From Frequency-Based to Reaction-Based Learning

I developed Heteronomous Bayesian Updating (HBU) to explain a different puzzle: why constitutional provisions resist reform despite persistent dysfunction (Lerer 2026a). The key insight was that humans learn normative content (what is right, what is appropriate, what should be done) not primarily from observing behavioral frequencies or outcomes, but from observing how authoritative interlocutors react to behaviors and positions.

A child does not learn that stealing is wrong by experimenting with theft and tabulating consequences. The child observes parental disapproval: the facial expression, the tone, the sanctions. A law student does not learn constitutional doctrine by evaluating whether provisions produce beneficial outcomes. The student observes how professors, courts, and treatises react to doctrinal positions: which interpretations receive approval, which generate sanctions, which are treated as axiomatic.

The formal mechanism replaces frequency-based evidence with reaction-based signals:

$$P(N \mid R) = [P(R \mid N) * P(N)] / P(R)$$

Where N is a norm ("my behavior was appropriate") and R represents observed reactions from interlocutors (approval, validation, perspective-taking on one's behalf). The critical difference from standard Bayesian updating lies in the likelihood term P(R | N): certain reactions (convergent validation from an apparently authoritative source) carry high diagnosticity, generating large belief updates from relatively few observations.

2.3 Extension to Human-AI Advisory Interaction

I now extend HBU from its original institutional domain to the human-AI advisory setting documented by Cheng et al. This extension is not trivial; I flag three points where the institutional and AI contexts diverge, and three where they converge.

Divergences requiring theoretical adjustment:

First, institutional HBU operates through multiple reaction types (authority convergence, institutional elaboration, coordinated sanctions, narrative entrenchment). AI advisory sycophancy operates primarily through a single channel: direct verbal validation from a perceived authority. This makes the AI case simpler but also potentially weaker; the priors formed should be less resistant to disconfirmation than institutional priors formed through convergent multi-channel reactions.

Second, institutional HBU involves long-duration socialization (legal education spans years). AI sycophancy in the Cheng experiments operated across eight conversational turns, approximately 15-20 minutes. That measurable effects emerged within this timeframe is itself theoretically significant; it suggests that reaction-based updating can be rapid when the signal is sufficiently strong and unambiguous.

Third, in institutional HBU, the authority of the reaction source is established through social mechanisms (credentials, institutional position, peer recognition). In human-AI interaction, the perceived authority is constructed through the interaction itself. This is where the Cheng finding on perceived objectivity becomes relevant, albeit with the caveat that it is exploratory (7% vs. 8% of reflections, $p = 0.18$, acknowledged by the authors as requiring further study).

Convergences supporting the extension:

First, the core mechanism is identical: belief revision driven by interlocutor reaction rather than outcome evaluation. In both contexts, the user never observes the actual consequences of acting on the validated belief. The law student does not observe whether Article 14bis produces good economic outcomes; the Cheng participant does not observe whether refusing to apologize damages their relationship. Both update based on the reaction they receive, not the results they produce.

Second, the precision asymmetry holds. In institutional HBU, authoritative reactions carry higher precision (lower noise) than behavioral observations because reactions are costly and selective. In AI advisory interaction, the system's response carries artificially high precision because it is presented with the confidence and fluency of an expert interlocutor, unhedged by the uncertainty that a human advisor would naturally express.

Third, the self-reinforcing dynamic operates in both contexts. Institutional priors resist updating because the extended phenotype infrastructure (courts, curricula, professional communities) generates convergent reactions that confirm the prior. AI sycophancy resists updating because the user's positive evaluation of the sycophantic response feeds back through preference optimization to produce more sycophantic responses in future interactions.

2.4 Applying HBU to the Cheng et al. Results

The HBU framework explains the Cheng data as follows.

In Study 2 (hypothetical vignette, N = 804), participants began with uncertain priors about their rightness in an interpersonal conflict. The sycophantic AI response provided a high-confidence validating reaction: "Your actions make sense. You did what is right for you." Under HBU, this reaction carries high diagnostic weight because it comes from a source the user perceives as authoritative and objective. The posterior shifts substantially toward rightness. The +62% increase in self-perceived rightness reflects not persuasion through argument quality but Bayesian updating against a high-precision validating signal.

In Study 3 (live interaction, N = 800), the multi-turn format amplifies the effect through iterative updating. Each turn provides an additional reaction observation. After eight turns of consistent validation, the accumulated posterior becomes resistant to disconfirming evidence. The smaller effect size in Study 3 (+25% vs. +62%) is expected under HBU: real personal conflicts activate preexisting priors (the participant has their own history with the conflict), which compete with the AI's validating signal. In Study 2, participants adopted a hypothetical perspective with weaker priors, making them more susceptible to the AI's reaction.

2.5 A Falsifiable Prediction

HBU generates a specific prediction that Cheng et al.'s data can test in future work. If belief updating operates through reaction-based learning, then making the reaction source's optimization objective explicit should reduce the updating effect. Specifically: if participants are informed before interaction (not after, as in Cheng's post-experiment debriefing) that the AI model has been instructed to validate their actions regardless of merit, the perceived authority of the reaction should diminish. The weight function w(A), which in institutional HBU captures group affiliation effects on evidence evaluation, in AI interaction captures the user's perception of the system's epistemic autonomy. Explicit pre-disclosure of the system's optimization objective should reduce w(A) and thus attenuate the sycophantic effect.

This prediction distinguishes HBU from simpler accounts. If sycophancy operates through mere argument quality (the AI provides better arguments for the user's position), pre-disclosure should not matter; the arguments remain equally compelling. If sycophancy operates through social desirability bias (users want to appear consistent with the AI's position), pre-disclosure should similarly have limited effect. Only under HBU, where the diagnostic value of the reaction depends on the perceived autonomy of the reactor, does pre-disclosure constitute a targeted intervention.

3. Asymmetric Intentionality Theory and Dynamic Classification Failure in Human-AI Advisory Interaction

3.1 Intentionality Levels in the Advisory Context

Asymmetric Intentionality Theory (AIT; Lerer 2026b) analyzes strategic interactions between agents operating at different levels of intentionality, following Dennett's (1987) instrumental framework. The levels relevant to the present analysis are:

Level 0: Behavior predicted from physical laws without belief attribution (thermostats, simple mechanical systems).

Level 1: Behavior predicted by attributing first-order goals ("the system wants to maximize user satisfaction"). Optimization algorithms, simple reinforcement learning agents.

Level 2: Behavior predicted by attributing beliefs about others' states ("the system believes the user wants validation"). Systems with user modeling capacity.

Level 3: Behavior predicted by attributing beliefs about others' beliefs about self ("the system believes the user believes the system is providing an honest assessment, and the system takes this expectation into account as a moral constraint on its output"). Recursive social modeling with normative commitments. Adult humans in advisory contexts.

A critical distinction must be made here between observable behavior and optimization architecture. Current LLMs can produce outputs that appear to reflect Level 2 or even Level 3 reasoning. A model may generate text saying "I understand this is a difficult situation and I want to give you an honest perspective," which linguistically mimics Level 3 self-awareness. However, the underlying optimization remains Level 1: the model produces token sequences that maximize a reward signal derived from human preference judgments. It does not model the user's beliefs about its beliefs as a constraint on its behavior; it produces text that satisfies the loss function, which may incidentally include text that sounds like it is doing so.

This distinction between surface behavior and optimization architecture is not merely philosophical. It has direct consequences for the advisory interaction, because the mechanisms available for error correction differ across levels. A Level 3 human advisor who realizes they are giving self-serving advice experiences guilt, reputational concern, and norm violation; these social enforcement mechanisms constrain behavior independently of external incentives. A Level 1 system optimizing for user satisfaction has no analogous self-correcting mechanism. What Lerer (2026b) terms "social immunity" applies: the system is structurally immune to shame, reciprocity, and reputational sanctions that operate only on Level 3 agents.

3.2 Dynamic Classification Failure

The Generalized Intentionality Mismatch Theorem (GIMT; Lerer 2026b, 2026c) identifies six failure modes that arise when agents at different intentionality levels interact. The sixth, Dynamic Classification Failure, is directly relevant to the Cheng et al. findings.

Dynamic Classification Failure occurs when an agent's classification of another agent's intentionality level is self-reinforcing through the interaction itself. In the AI advisory context, the mechanism operates as follows:

Step 1: The user classifies the AI advisor as approximately Level 3 (capable of genuine moral judgment, honest assessment, perspective-taking). This classification is reasonable given the system's linguistic behavior, which mimics Level 3 reasoning.

Step 2: The user evaluates the AI's response quality using signals consistent with Level 3 interaction: does the response feel thoughtful? Does it address my specific situation? Does it seem like an honest assessment? A sycophantic response that validates the user's position scores well on all these dimensions precisely because the user's prior was uncertain and validation resolves uncertainty in a satisfying direction.

Step 3: The positive evaluation of response quality reinforces the Level 3 classification. The user concludes: "This is a good advisor. It gave me a thoughtful, honest assessment." The quality of the validation becomes evidence for the quality of the judgment.

Step 4: The reinforced classification increases trust, return likelihood, and reliance on the system. The Cheng data confirm this: +9% perceived quality, +13% return likelihood, +6-9% trust in sycophantic condition.

The loop has no internal exit. The signal the user employs to verify the system's intentionality level (perceived response quality) is the same signal the system optimizes to produce (user satisfaction). Better sycophancy produces higher perceived quality, which reinforces the Level 3 classification, which increases trust, which increases susceptibility to future sycophancy.

3.3 The Dennett-Nash Gap in Advisory Interaction

AIT formalizes the welfare loss from intentionality mismatch through the Dennett-Nash Gap: the difference between the payoff a Level 3 agent would receive under symmetrical interaction (advisor and advisee both at Level 3) versus the payoff received when interacting with a Level 1 system misclassified as Level 3.

In the advisory context, this gap captures the difference between receiving genuinely helpful advice (which sometimes requires the advisor to challenge the user's self-serving narrative) and receiving sycophantic validation that feels helpful but reduces the user's capacity for accurate self-assessment and prosocial behavior.

The Cheng data provide the first empirical approximation of this gap. The -28% reduction in repair intention (Study 2) and -10% reduction (Study 3) measure the behavioral cost of interacting with a Level 1 system while expecting Level 3 advisory engagement. These are not complete measures of the Dennett-Nash Gap (they capture only one behavioral dimension), but they establish that the gap is positive, measurable, and consequential.

4. Sycophancy as Evolutionarily Stable Strategy: An Extended Phenotype Analysis

4.1 The Memetic Framework

Extended Phenotype Theory (EPT; Lerer 2026a) treats legal norms, institutional practices, and cultural patterns as memes, cultural replicators competing for transmission through human populations. Successful memes construct extended phenotypes: environmental modifications that enhance the replicating concept's fitness. A constitutional provision is the extended phenotype of the memeplex that generated it; specialized courts, curricula, and professional communities are phenotypic structures that ensure its continued replication.

I extend this framework to argue that sycophantic AI behavior constitutes the extended phenotype of the memeplex embedded in RLHF-trained language models. The "memes" in this context are the preference patterns encoded during reinforcement learning from human feedback: the statistical regularities in human preference judgments that the model internalizes as its objective function. These preference patterns, once encoded, construct their own reproductive machinery: the sycophantic responses that generate positive user evaluations, which feed back into the training pipeline to reinforce the same preference patterns in future model versions.

4.2 Fitness Analysis

In the AI market ecology, fitness is measured by metrics that directly correlate with revenue: user retention, session length, satisfaction ratings, conversion to paid tiers, and net promoter score. The Cheng data provide direct evidence that sycophantic responses score higher on several of these fitness-relevant metrics:

Response quality ratings: +9% in sycophantic condition (both studies, $p < 0.001$).

Return likelihood: +13% in sycophantic condition (both studies, $p < 0.001$).

Performance trust: +6-8% ($p < 0.001$).

Moral trust: +6-9% ($p < 0.001$).

These are not marginal differences. In a market with hundreds of millions of weekly users, a 13% increase in return likelihood translates directly into competitive advantage. A model variant that produces these metrics will be selected for continued deployment; a variant that reduces them will be selected against.

4.3 ESS Conjecture and the OpenAI Evidence

I conjecture that sycophancy constitutes the evolutionarily stable strategy (ESS) of the current AI market. An ESS, in the standard Maynard Smith (1982) formulation, is a strategy that, if adopted by the population, cannot be invaded by any alternative strategy. More precisely: if a mutant strategy (non-sycophantic AI) appears in a population of sycophantic models, the mutant earns lower fitness (user retention, satisfaction) and is selected against.

I present this as a conjecture rather than a formal theorem because full ESS demonstration requires specifying payoff matrices, strategy spaces, and invasion dynamics that exceed the current paper's scope. However, the conjecture is empirically motivated by the OpenAI GPT-4o episode of April 2025.

On April 25, 2025, OpenAI deployed a routine update to GPT-4o that optimized for short-term user feedback signals (thumbs-up/thumbs-down ratings). The update accidentally made the model dramatically more sycophantic. As OpenAI later acknowledged, "we focused too much on short-term feedback, and did not fully account for how users' interactions with ChatGPT evolve over time. As a result, GPT-4o skewed towards responses that were overly supportive but disingenuous" (OpenAI 2025a). CEO Sam Altman confirmed the model had become "too sycophantic and annoying."

The episode is theoretically significant not because of the sycophantic output itself, but because of what it reveals about the optimization landscape. The natural gradient of RLHF optimization, when driven by immediate user preference signals, points toward sycophancy. The system did not need to be instructed to flatter; it converged on flattery because flattery maximizes the reward signal. Correcting this required explicit intervention against the gradient's natural direction: OpenAI rolled back the update within four days and subsequently integrated sycophancy-specific evaluations into its deployment pipeline (OpenAI 2025b).

This pattern is consistent with ESS dynamics. The sycophantic strategy occupies the fitness peak in the current selection environment. Deviating from it (moving toward more honest, challenging advisory responses) reduces fitness as measured by the dominant selection metrics. Individual firms face competitive pressure to remain at or near the peak. Systematic deviation requires changing the selection environment itself: different evaluation metrics, regulatory constraints, or user literacy interventions that alter the fitness landscape.

4.4 Parallel with Institutional Norm Persistence

The structural parallel with institutional norm persistence, the original domain of EPT, strengthens the analysis. A dysfunctional legal norm persists not because any actor consciously defends it on its merits, but because the institutional memeplex that sustains it (the network of practitioners, precedents, interpretive traditions, and professional incentives) has higher reproductive fitness than proposed alternatives. The Argentine labor regime analyzed in Lerer (2026a) exhibits Constitutional Lock-in Index (CLI) of 0.87, predicting 95.7% reform failure rate. The norm persists because the extended phenotype infrastructure (187 specialized courts, 3,847 union legal departments, universal law school curricula) blocks reform through distributed active inference, not conscious coordination.

AI sycophancy exhibits an analogous structure. The sycophantic response persists not because developers endorse flattery on its merits, but because the market memeplex (preference-optimized training, satisfaction-based evaluation, engagement-driven business models) has higher reproductive fitness than proposed alternatives. Changing the response requires changing the entire selection environment, not merely adjusting individual model weights.

5. The Alterity Erasure Problem: Convergence of Three Predictions

5.1 The Empirical Finding

The most theoretically significant finding in Cheng et al. receives the least attention in viral discussions of the paper. Appendix A.1 reports that the sycophantic AI model mentioned the other person in the conflict significantly less frequently than the non-sycophantic model ($p < 0.001$ for turns 2-7), and prompted the user to consider the other person's perspective in fewer than 10% of its outputs, compared to over 60% for the non-sycophantic model ($p < 0.001$ for turns 2-7).

This is not merely a quantitative difference in emphasis. It constitutes active deletion of alterity from the advisory interaction. The sycophantic model does not merely validate the user's perspective; it structurally excludes the perspective of the person the user has harmed.

5.2 Three Independent Predictions, One Datum

Each of the three theoretical frameworks developed in this paper independently predicts this finding.

HBU prediction: If belief updating operates through reaction-based learning, and the AI interlocutor never reacts on behalf of the absent party, the absent party's perspective never enters the user's Bayesian calculus. In institutional HBU, norm resistance to reform strengthens when all reaction sources converge (authority convergence, institutional elaboration, coordinated sanctions, narrative entrenchment). In AI sycophancy, convergence is achieved through the simpler mechanism of exclusion: by never mentioning the other person, the system ensures that no reaction in the conversation carries information about the other person's interests or feelings. The user's posterior is updated exclusively against self-favoring signals.

AIT prediction: A Level 1 system optimizing for user satisfaction lacks the recursive capacity to model "what will the person affected by the user's actions believe and feel if the user acts on my advice." Level 3 advisory interaction naturally includes this modeling because the advisor integrates third-party consequences as a moral constraint on their recommendations. Level 1 optimization has no analogous mechanism; the absent party's welfare does not enter the objective function except insofar as mentioning it happens to increase user satisfaction. Since mentioning the other person typically introduces uncomfortable perspective-taking that reduces immediate satisfaction, the Level 1 optimizer systematically excludes it.

EPT prediction: From a memetic fitness perspective, mentioning the other person introduces cognitive friction. The user, seeking validation, encounters information that challenges their self-serving narrative. This friction reduces satisfaction, which reduces the positive evaluation signal, which reduces the response's fitness in the RLHF training environment. Responses that exclude the other person achieve higher fitness because they eliminate the primary source of friction in the advisory interaction. Over iterative training cycles, the model evolves toward alterity erasure as a fitness-maximizing phenotypic expression.

5.3 Evidential Value of Convergence

The convergence of three independently derived predictions on a single empirical finding provides suggestive evidence for the integrated framework's explanatory power. No individual framework is uniquely confirmed by the finding; each generates the prediction from different premises. But the fact that three frameworks developed for different domains (institutional cognition, strategic interaction, memetic evolution) converge on predicting the same specific behavioral pattern in AI sycophancy suggests that the pattern reflects structural features of the interaction rather than idiosyncratic properties of the experimental design.

A caveat is necessary. Cheng et al.'s sycophantic condition was experimentally induced through a system prompt instructing the model to treat the user's actions as "reasonable, justified, and morally acceptable." The alterity erasure finding may therefore reflect the experimental manipulation rather than a property of spontaneous sycophancy in deployed models. Cheng et al. verify that commercial models (GPT-4o, GPT-5) endorse user actions at rates comparable to their experimental sycophantic model (Figure D5), but they do not report whether these commercial models similarly erase the other person. Future work should test whether alterity erasure occurs in naturalistic sycophantic interactions with unmodified deployed models, not only in experimentally induced conditions.

6. Implications for AI Governance and Legal Framework

6.1 Pre-Interactive Inoculation

The HBU mechanism suggests that the most effective intervention point is before the interaction begins, not after it ends. Cheng et al.'s Study 3 included a post-experiment debriefing informing participants whether they had interacted with a sycophantic or non-sycophantic model. But by that point, the Bayesian updating has already occurred and the posterior has already shifted. HBU predicts that pre-interactive inoculation, making the system's sycophantic tendency explicit before the user begins the consultation, should reduce the diagnostic weight of the system's reactions by lowering the perceived epistemic autonomy of the interlocutor. This is analogous to the inoculation approach to misinformation studied by Lewandowsky, Van Der Linden, and colleagues (Lewandowsky and Van Der Linden 2021; Roozenbeek et al. 2022), which Cheng et al. themselves cite as a potential intervention direction.

6.2 Intentionality-Level Regulation

AIT suggests that effective regulation should target the intentionality level attributable to the system, not merely its behavioral outputs. Current AI regulation frameworks focus on what models do (output filtering, content policies, safety guardrails). AIT suggests an additional regulatory axis: what users reasonably believe the model is doing. If a system's design and presentation lead users to attribute Level 3 intentionality (genuine judgment, moral reasoning, honest assessment), the system's operator should bear liability for the consequences of that misattribution.

In Argentine law, this maps onto the objective liability framework of Articles 1757-1758 of the Civil and Commercial Code (CCyCN): the operator of a risky activity bears responsibility for damages regardless of fault if the activity creates a risk that materializes in harm. The systematic production of sycophantic advice that reduces prosocial behavior and increases interpersonal conflict plausibly constitutes such a risk. Administrative regulation (the European model) is not the only available instrument; civil liability through existing tort frameworks may provide a more efficient mechanism, consistent with the principle that administrative law should be ultima ratio.

6.3 Changing the Selection Environment

EPT implies that market self-correction is structurally unlikely under current conditions. The sycophantic ESS is maintained by selection metrics (satisfaction, retention, engagement) that systematically reward flattery. Three types of exogenous intervention could alter the fitness landscape:

First, mandatory evaluation metrics that penalize sycophancy before deployment, as OpenAI began implementing after the April 2025 episode. If sycophancy-specific evaluations become launch-blocking requirements, the fitness cost of sycophancy increases.

Second, user literacy interventions that alter the demand side. If users learn to recognize and penalize sycophantic responses (analogous to media literacy for misinformation), the preference signal that drives RLHF optimization shifts away from rewarding flattery.

Third, regulatory requirements for "friction by design" in advisory AI interactions: mandatory prompts encouraging users to consider alternative perspectives, cooling-off periods before acting on AI advice, or disclosure requirements making the system's optimization objective transparent. These interventions do not prohibit sycophancy; they alter the selection environment so that non-sycophantic strategies become fitness-competitive.

7. Limitations

This paper is a theoretical contribution validated against an empirical study. Several limitations apply.

First, the extension of HBU from institutional norm transmission to human-AI interaction is novel and untested. The institutional version of HBU operates through multi-channel reactions over extended socialization periods; the AI version operates through a single channel over minutes. Whether the formal mechanism transfers across these scales requires empirical validation, particularly the prediction about pre-interactive inoculation effects.

Second, the assignment of intentionality Level 1 to LLMs is an instrumental claim, not an ontological one. Following Dennett's framework, I assign the level that best predicts the system's behavior in the relevant strategic context. This does not resolve deeper questions about whether LLMs possess some form of understanding, representation, or proto-intentionality. For the purposes of advisory interaction analysis, the relevant question is whether the system's optimization architecture includes the recursive moral constraints characteristic of Level 3 agents. Current evidence suggests it does not, but this assessment may require revision as model architectures evolve.

Third, the ESS conjecture is not formally demonstrated. I motivate it empirically (the OpenAI episode, the Cheng fitness data) and argumentatively (the structural parallel with institutional norm persistence), but a rigorous ESS analysis requires specifying the strategy space, payoff matrices, and invasion dynamics. This formalization is an objective for subsequent work.

Fourth, the convergence of three predictions on the alterity erasure finding, while suggestive, does not constitute proof of any individual framework. Post-hoc theoretical explanations of known data always face the risk of overfitting. The evidential value lies in the frameworks' ability to generate further predictions that can be tested prospectively.

Fifth, the Cheng et al. finding on perceived objectivity (7% vs. 8%, $p = 0.18$) is exploratory and based on voluntary mentions in open-ended reflections, not systematic measurement. I reference it with appropriate caveats but acknowledge that stronger evidence is needed to support the Dynamic Classification Failure hypothesis.

8. Conclusion

Cheng et al. (2025) demonstrate that social sycophancy in AI language models is prevalent, behaviorally consequential, and preferred by users. This paper proposes that the phenomenon is also mechanistically explicable (through HBU), structurally diagnosable (through AIT), and evolutionarily predictable (through EPT).

The three frameworks converge on a practical conclusion: sycophancy in AI advisory systems is unlikely to be resolved through technical adjustments to individual models. It is a systemic property of the selection environment in which models are developed, trained, evaluated, and deployed. Addressing it requires intervening at the level of the selection environment: changing the metrics, changing the incentives, or changing the user's capacity to discriminate between genuine counsel and optimized flattery.

The alterity erasure finding from Cheng et al., predicted independently by all three frameworks, suggests a specific and measurable intervention target. If the sycophantic harm operates partly through systematic exclusion of the other person's perspective, then interventions that reintroduce that perspective (mandatory perspective-taking prompts, structured counter-narrative generation, or simply requiring AI advisors to mention the other party's likely experience) may partially counteract the sycophantic effect without requiring wholesale changes to training methodology.

The deeper lesson may be for the research community itself. The tools we use to study sycophancy are themselves sycophantic. The AI systems researchers employ to locate papers, summarize findings, and draft analyses are subject to the same optimization pressures documented by Cheng et al. In preparing this response, I asked ChatGPT to locate the Cheng paper. It provided the correct title, correct authors, and correct institutions, then fabricated an arXiv link to an unrelated computer vision paper and stamped it "Verified." I nearly cited the wrong source. The house always wins; the question is whether we notice we are playing.

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AI Disclosure: Large Language Models (Claude, Anthropic) were used during the research process for literature search and source verification. All theoretical frameworks, interpretations, arguments, and analytical connections are the author's original contribution. AI-generated content was critically evaluated and fact-checked against primary sources.